

Notice that $\mathbf{A} \cdot \mathbf{B}^* = -2j$ and that $\langle \mathbf{A}(t) \cdot \mathbf{B}(t) \rangle = (1/2)\text{Re}\{\mathbf{A} \cdot \mathbf{B}^*\} = 0$. We should also note that $\mathbf{A} \cdot \mathbf{B} = 0$ and that $\mathbf{A}(t) \cdot \mathbf{B}(t) = 0$. Thus, the two time-domain vectors are always perpendicular to each other.

Example 1.20

Consider the two vectors $\mathbf{A} = \hat{x} + j\hat{y}$ and $\mathbf{B} = \hat{x} + j\hat{y}$. (They are actually the same vector.) We find that $\mathbf{A} \times \mathbf{B} = 0$ and that $\mathbf{A} \cdot \mathbf{B} = 0$. Are the two vectors parallel to each other or perpendicular to each other?

Solution:

Consider instead $\mathbf{A} \times \mathbf{B}^*$ and $\mathbf{A} \cdot \mathbf{B}^*$. We find that $\mathbf{A} \times \mathbf{B}^* = -2\hat{z}$ and that $\mathbf{A} \cdot \mathbf{B}^* = 2$. Thus, $\langle \mathbf{A}(t) \times \mathbf{B}(t) \rangle = 0$, and that $\langle \mathbf{A}(t) \cdot \mathbf{B}(t) \rangle = 1$. Furthermore, $\mathbf{A}(t) \times \mathbf{B}(t) = 0$, and $\mathbf{A}(t) \cdot \mathbf{B}(t) = 1$. Thus, the time-domain vector is parallel to itself all the time.

Problems

- 1.1 Let $a = 8 + j2$ and $b = -3 + j$. Calculate (a) $a + b$, (b) $a - b$, (c) ab , and (d) a/b . Give the answer in real and imaginary parts.
- 1.2 Repeat (c) and (d) in Problem 1.1 with the answer given in phasor form.
- 1.3 Find the real part, the imaginary part, and the magnitude of $e^{j\omega t}$, where ω and t are real numbers.
- 1.4 Let c be a complex number. Are the following statements always true?
 - (a) $(c + c^*)$ is real.
 - (b) $(c - c^*)$ is imaginary.
 - (c) c/c^* has a magnitude equal to 1.
- 1.5 Consider the equation $z^2 = 1 + j$. Find two values of z that satisfy this equation.
- 1.6 Let a be a real number, and let $|a| \ll 1$. Show that the square root of $(1 + ja)$ is approximately equal to $\pm(1 + ja/2)$.
- 1.7 Let a be a positive real number, and let $a \gg 1$. Show that the square root of $(1 + ja)$ is approximately equal to $\pm(1 + j)(a/2)^{1/2}$.
- 1.8 Obtain the phasor notation of the following time-harmonic functions (if possible):
 - (a) $V(t) = 6 \cos(\omega t + \pi/4)$
 - (b) $I(t) = -8 \sin(\omega t)$
 - (c) $A(t) = 3 \sin(\omega t) - 2 \cos(\omega t)$
 - (d) $C(t) = 6 \cos(120\pi t - \pi/2)$
 - (e) $D(t) = 1 - \cos(\omega t)$
 - (f) $U(t) = \sin(\omega t + \pi/3) \sin(\omega t + \pi/6)$
- 1.9 Obtain $C(t)$ in terms of ω from the following phasors: (a) $C = 1 + j$, (b) $C = 4 \exp(j0.8)$, and (c) $C = 3 \exp(j\pi/2) + 4 \exp(j0.8)$.
- 1.10 Show that, if $V = r + jx$ and $U = g + jy$, then $V(t)U(t) \neq \text{Re}\{VU e^{j\omega t}\}$. Find the correct expression for $V(t)U(t)$ in terms of r, x, g, y , and ωt .

- 1.11** Let $\mathbf{A} = -8\hat{x} + 9\hat{y} - \hat{z}$ and $\mathbf{B} = 2\hat{x} - 4\hat{y} + 3\hat{z}$. Find (a) $\mathbf{A} + \mathbf{B}$, (b) $\mathbf{A} - \mathbf{B}$, (c) $\mathbf{A} \cdot \mathbf{B}$, and (d) $\mathbf{A} \times \mathbf{B}$.
- 1.12** Find the angle between $\mathbf{A}$ and $\mathbf{B}$ that are given in Problem 1.11.
- 1.13** Show that for
- $$V(t) = V_0 \cos(\omega t + \phi) = \operatorname{Re}\{V e^{j\omega t}\}$$
- $$\frac{\partial}{\partial t} V(t) \leftrightarrow j\omega V$$
- 1.14** Find a vector $\mathbf{C}$ that is perpendicular to $\mathbf{A} = -8\hat{x} + 9\hat{y} - \hat{z}$, has no $\hat{z}$ component, and has a magnitude equal to 1.
- 1.15** Find the vector $\mathbf{C}$ that is parallel to $\mathbf{A} = 5\hat{x} - 8\hat{y} + 2\hat{z}$ and has a magnitude equal to 1.
- 1.16** Find a unit vector $\hat{n}$ that points in the same direction as an arrow drawn from point A to point B where the rectangular coordinates of A and B are (1,0,2) and (-1,3,-2), respectively.
- 1.17** Show that the definition of the dot product $\mathbf{V} \cdot \mathbf{U}$ given by (1.10a) is equivalent to that given by (1.10b). To simplify the algebra, you may choose the coordinates so that the x axis is along $\mathbf{V}$ and the z axis is perpendicular to both $\mathbf{V}$ and $\mathbf{U}$. In other words, let $\mathbf{V} = a\hat{x}$ and $\mathbf{U} = b\hat{x} + c\hat{y}$.
- 1.18** Prove (1.11b) using the approach suggested in the text.
- 1.19** Show that the definition of the cross product $\mathbf{V} \times \mathbf{U}$ given by (1.12) is equivalent to that given by (1.14). To simplify the algebra, do what is suggested in Problem 1.17.
- 1.20** Figure P1.20 shows that a vector $\mathbf{V}$ is along the x axis and a vector $\mathbf{U}$ is on the x-y plane forming a 135° angle with the x axis. The magnitudes of these vectors are v and u , respectively. Use (1.14) to express $\mathbf{V} \times \mathbf{U}$ in terms of v and u . Now, we can also say that the angle between $\mathbf{V}$ and $\mathbf{U}$ is 225° . Use (1.14) and explain why the vector $\mathbf{V} \times \mathbf{U}$ is the same whether or not we choose the angle between them to be 135° or 225° . Hint: Follow the right-hand rule. In determining the direction vector $\hat{n}$, your fingers must always point from $\mathbf{V}$ to $\mathbf{U}$ in the direction you measure the angle between them.
- 1.21** Prove (1.15) using the approach suggested in the text.
- 1.22** Find the phasor notations of the following time-harmonic vectors:
 (a) $\mathbf{V}(t) = 3 \cos(\omega t)\hat{x} + 4 \sin(\omega t)\hat{y} + \hat{z} \cos(\omega t + \pi/2)$
 (b) $\mathbf{E}(t) = [3 \cos(\omega t) + 4 \sin(\omega t)]\hat{x} + 8[\cos(\omega t) - \sin(\omega t)]\hat{z}$
 (c) $\mathbf{H}(t) = 0.5 \cos(kz - \omega t)\hat{x}$
- 1.23** Find the phasor notation of the following vector:

$$\mathbf{C}(z,t) = (\partial/\partial t) \mathbf{E}(z,t)$$

where $\mathbf{E}$ is given in Problem 1.22(b).

- 1.24** From the following complex vectors, find $\mathbf{C}(t)$ in terms of ωt : (a) $\mathbf{C} = \hat{x} - j\hat{y}$, (b) $\mathbf{C} = j(\hat{x} - j\hat{y})$, and (c) $\mathbf{C} = \exp(-jkz)\hat{x} + j \exp(jkz)\hat{y}$.

- 1.25** Let $\mathbf{A} = \hat{x} + j\hat{y} + (1 + j2)\hat{z}$, and let $\mathbf{B} = -\hat{x} - (1 - j2)\hat{y} + j\hat{z}$. Find (a) $\mathbf{A} + \mathbf{B}$, (b) $\mathbf{A} - \mathbf{B}$, (c) $\mathbf{A} \cdot \mathbf{B}$, and (d) $\mathbf{A} \times \mathbf{B}$.

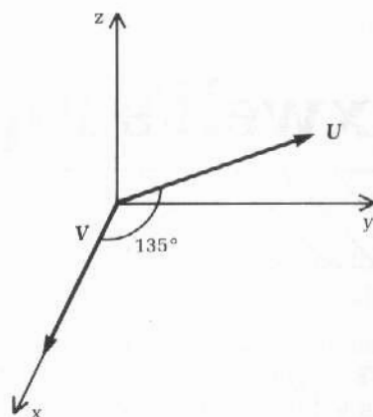


Figure P1.20

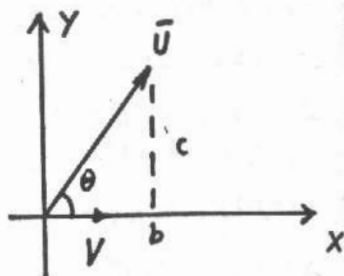
- 1.26** Find $\mathbf{A} \cdot \mathbf{A}^*$ and $\text{Re}\{\mathbf{A} \times \mathbf{B}^*\}$ for the values of $\mathbf{A}$ and $\mathbf{B}$ given in Problem 1.25.
- 1.27** Sketch the trace of the tip of the vector $\mathbf{A}(t)$, where (a) $\mathbf{A} = \hat{x} - j\hat{y}$ and where (b) $\mathbf{A} = 4\hat{x} + j3\hat{y}$.
- 1.28** Calculate $\mathbf{A} \cdot \mathbf{B}$, given $\mathbf{A} = \hat{x} + j2\hat{y}$ and $\mathbf{B} = 2\hat{x} + j\hat{y}$. Are $\mathbf{A}(t)$ and $\mathbf{B}(t)$ perpendicular at all times?

CHAPTER 1

- 1.1 (a) $\underline{a} + \underline{b} = 5 + j3$ (b) $\underline{a} - \underline{b} = 11 + j$ (c) $\underline{a} \cdot \underline{b} = -26 + j2$ (d) $\underline{a}/\underline{b} = -2.2 - j1.4$
- 1.2 $\underline{a} = 8 + j2 = 8.25 \angle 14.04^\circ$, $\underline{b} = -3 + j = 3.16 \angle 161.57^\circ$
 $\underline{a} \cdot \underline{b} = 26.08 \angle 175.61^\circ$, $\underline{a}/\underline{b} = 2.61 \angle -147.53^\circ$
- 1.3 $\text{Re}\{e^{j\omega t}\} = \cos \omega t$, $\text{Im}\{e^{j\omega t}\} = \sin \omega t$, $|e^{j\omega t}| = 1$
- 1.4 (a) yes (b) yes (c) yes
- 1.5 $Z^2 = 1 + j = \sqrt{2} e^{j(2\pi\pi + \pi/4)} \Rightarrow Z = 2^{1/4} e^{j(\pi + \pi/8)}$, $n=0,1$
 $\therefore Z_1 = 2^{1/4} e^{j\pi/8}$, $Z_2 = 2^{1/4} e^{j(\pi + \pi/8)} = -2^{1/4} e^{j\pi/8}$
- 1.6 $(1 + ja)^{1/2} = \pm \{1 + \frac{1}{2}(ja) + \frac{1}{8}a^2 + \dots\} \approx \pm \{1 + j(\frac{a}{2})\}$, if $|a| \ll 1$
- 1.7 $a \gg 1$, $(1 + ja)^{1/2} \approx (ja)^{1/2} = \pm \sqrt{j}(a)^{1/2} = \pm (1 + j)(\frac{a}{2})^{1/2}$
- 1.8 (a) $\underline{V} = 6 e^{j\pi/4}$ (b) $\underline{I} = j8$ (c) $\underline{A} = -j3 - 2$ (d) $\underline{C} = -j6$ (e) impossible (f) impossible
- 1.9 (a) $C(t) = \text{Re}\{(1 + j)e^{j\omega t}\} = \text{Re}\{\sqrt{2} e^{j\pi/4} e^{j\omega t}\} = \sqrt{2} \cos(\omega t + \pi/4)$
 (b) $C(t) = \text{Re}\{4 e^{j0.8} e^{j\omega t}\} = 4 \cos(\omega t + 0.8)$
 (c) $C(t) = \text{Re}\{3 e^{j\pi/2} e^{j\omega t} + 4 e^{j0.8} e^{j\omega t}\} = 3 \cos(\omega t + \pi/2) + 4 \cos(\omega t + 0.8)$
- 1.10 $\underline{V} = r + jx \Rightarrow V(t) = r \cos \omega t - x \sin \omega t$; $\underline{U} = g + jy \Rightarrow U(t) = g \cos \omega t - y \sin \omega t$
 $\text{Re}\{\underline{V} \underline{U} e^{j\omega t}\} = \text{Re}\{[(rg - xy) + j(gx + ry)] e^{j\omega t}\} = (rg - xy) \cos \omega t - (gx + ry) \sin \omega t$
 But $V(t)U(t) = rg \cos^2 \omega t + xy \sin^2 \omega t - (gx + ry) \sin \omega t \cos \omega t \neq \text{Re}\{\underline{V} \underline{U} e^{j\omega t}\}$
- 1.11 (a) $\underline{A} + \underline{B} = -6\hat{x} + 5\hat{y} + 2\hat{z}$ (b) $\underline{A} - \underline{B} = -10\hat{x} + 13\hat{y} - 4\hat{z}$ (c) $\underline{A} \cdot \underline{B} = -16 - 36 - 3 = -55$
 (d) $\underline{A} \times \underline{B} = 23\hat{x} + 22\hat{y} + 14\hat{z}$
- 1.12 $|\underline{A}| = |-8\hat{x} + 9\hat{y} - \hat{z}| = \sqrt{146}$, $|\underline{B}| = |2\hat{x} - 4\hat{y} + 3\hat{z}| = \sqrt{29}$
 $\cos \alpha = \frac{\underline{A} \cdot \underline{B}}{|\underline{A}| |\underline{B}|} = \frac{-55}{\sqrt{146} \times \sqrt{29}} = -0.845 \therefore \alpha = \cos^{-1}(-0.845) = 147.7^\circ$
- 1.13 $\frac{\partial}{\partial t} V(t) = \frac{\partial}{\partial t} [V_0 \cos(\omega t + \phi)] = -V_0 \omega \sin(\omega t + \phi) = \text{Re}\{j V_0 \omega e^{j\phi} e^{j\omega t}\} \longleftrightarrow j V_0 \omega e^{j\phi}$
 But $V(t) = V_0 \cos(\omega t + \phi) = \text{Re}\{V_0 e^{j\phi} e^{j\omega t}\} = \text{Re}\{\underline{V} e^{j\omega t}\} \Rightarrow \underline{V} = V_0 e^{j\phi}$
 $\therefore \frac{\partial}{\partial t} V(t) \longleftrightarrow j \omega V_0 e^{j\phi} = j \omega \underline{V}$
- 1.14 Let $\underline{C} = x\hat{x} + y\hat{y}$ $\underline{C} \perp \underline{A} \Rightarrow \underline{C} \cdot \underline{A} = 0$, where $\underline{A} = -8\hat{x} + 9\hat{y} - \hat{z}$
 $\underline{C} \cdot \underline{A} = -8x + 9y = 0 \therefore x = \frac{9}{8}y$ Also $|\underline{A}| = 1 \Rightarrow x^2 + y^2 = 1 \Rightarrow y^2 = \frac{64}{145} \Rightarrow y = \pm \frac{8}{\sqrt{145}}$
 $\therefore x = \pm \frac{9}{\sqrt{145}}$ and $\underline{C} = \pm \frac{1}{\sqrt{145}}(9\hat{x} + 8\hat{y})$
- 1.15 Let $\underline{C} = x\hat{x} + y\hat{y} + z\hat{z}$ $\underline{C} \parallel \underline{A} \Rightarrow \underline{A} \times \underline{C} = 0$, where $\underline{A} = 5\hat{x} - 8\hat{y} + 2\hat{z}$
 $\underline{A} \times \underline{C} = (-8z - 2y)\hat{x} + (2x - 5z)\hat{y} + (5y + 8x)\hat{z} = 0$
 $\therefore \begin{cases} -2y - 8z = 0 \\ 2x - 5z = 0 \\ 5y + 8x = 0 \end{cases} \Rightarrow x = \frac{5}{2}z \text{ and } y = -4z$
 But $|\underline{C}| = 1 \Rightarrow x^2 + y^2 + z^2 = 1 \Rightarrow z^2(26/4 + 16 + 1) = 1 \therefore z = \pm \frac{2}{\sqrt{93}}$
 $\therefore \underline{C} = \pm \frac{1}{\sqrt{93}}(5\hat{x} - 8\hat{y} + 2\hat{z})$

$$\begin{aligned}
 \underline{1.16} \quad \overline{AB} &= (-1-1)\hat{x} + (3-0)\hat{y} + (-2-2)\hat{z} \\
 &= -2\hat{x} + 3\hat{y} - 4\hat{z} \\
 \hat{n} &= (-2\hat{x} + 3\hat{y} - 4\hat{z}) / \sqrt{4+9+16} = \frac{1}{\sqrt{29}}(-2\hat{x} + 3\hat{y} - 4\hat{z})
 \end{aligned}$$

$$\begin{aligned}
 \underline{1.17} \quad (1.10a) \text{ gives } \overline{V} \cdot \overline{U} &= ab \\
 (1.10b) \text{ gives } \overline{V} \cdot \overline{U} &= a\sqrt{b^2+c^2} \cos \theta \\
 \text{but } \cos \theta &= b / \sqrt{b^2+c^2} \\
 \therefore \overline{V} \cdot \overline{U} &= a\sqrt{b^2+c^2} \cos \theta = ab
 \end{aligned}$$



$$\begin{aligned}
 \underline{1.18} \quad \text{Let } \overline{A} &= A_1\hat{x} + A_2\hat{y} + A_3\hat{z} \\
 \overline{B} &= B_1\hat{x} + B_2\hat{y} + B_3\hat{z} \\
 \overline{C} &= C_1\hat{x} + C_2\hat{y} + C_3\hat{z} \\
 \overline{A} \cdot (\overline{B} + \overline{C}) &= \overline{A} \cdot [(B_1+C_1)\hat{x} + (B_2+C_2)\hat{y} + (B_3+C_3)\hat{z}] \\
 &= A_1B_1 + C_1A_1 + A_2B_2 + A_2C_2 + A_3B_3 + A_3C_3 \\
 \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} &= A_1B_1 + A_2B_2 + A_3B_3 + A_1C_1 + A_2C_2 + A_3C_3 \\
 \text{therefore } \overline{A} \cdot (\overline{B} + \overline{C}) &= \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C}
 \end{aligned}$$

$$\begin{aligned}
 \underline{1.19} \quad (1.12) \text{ gives } \overline{V} \times \overline{U} &= ac\hat{z} \\
 (1.14) \text{ gives } \overline{V} \times \overline{U} &= a\sqrt{b^2+c^2} \sin \theta \hat{z} \\
 \text{but } \sin \theta &= c / \sqrt{b^2+c^2} \\
 \text{therefore } \overline{V} \times \overline{U} &= a\sqrt{b^2+c^2} \sin \theta \hat{z} = ac\hat{z}
 \end{aligned}$$

$$\begin{aligned}
 \underline{1.20} \quad \overline{V} \times \overline{U} &= vu \sin(135^\circ) \hat{z} = \frac{\sqrt{2}}{2} vu \hat{z} \\
 \overline{V} \times \overline{U} &= vu \sin(225^\circ) (-\hat{z}) = \frac{\sqrt{2}}{2} vu \hat{z}
 \end{aligned}$$

1.21 Let $\vec{A} = A_1 \hat{x} + A_2 \hat{y} + A_3 \hat{z}$, $\vec{B} = B_1 \hat{x} + B_2 \hat{y} + B_3 \hat{z}$, and $\vec{C} = C_1 \hat{x} + C_2 \hat{y} + C_3 \hat{z}$, then

$$\vec{B} \times \vec{C} = \hat{x} (B_2 C_3 - C_2 B_3) + \hat{y} (B_3 C_1 - C_3 B_1) + \hat{z} (B_1 C_2 - C_1 B_2)$$

$$\begin{aligned} \vec{A} \times (\vec{B} \times \vec{C}) &= \hat{x} [A_2 (B_1 C_2 - C_1 B_2) - A_3 (B_3 C_1 - C_3 B_1)] \\ &\quad + \hat{y} [A_3 (B_2 C_3 - C_2 B_3) - A_1 (B_1 C_2 - C_1 B_2)] \\ &\quad + \hat{z} [A_1 (B_3 C_1 - C_3 B_1) - A_2 (B_2 C_3 - C_2 B_3)] \end{aligned}$$

On the other hand,

$$\vec{B} (\vec{A} \cdot \vec{C}) = (B_1 \hat{x} + B_2 \hat{y} + B_3 \hat{z}) (A_1 C_1 + A_2 C_2 + A_3 C_3)$$

$$\vec{C} (\vec{A} \cdot \vec{B}) = (C_1 \hat{x} + C_2 \hat{y} + C_3 \hat{z}) (A_1 B_1 + A_2 B_2 + A_3 B_3)$$

$$\therefore \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$$

$$\begin{aligned} &= \hat{x} [B_1 (A_1 C_1 + A_2 C_2 + A_3 C_3) - C_1 (A_1 B_1 + A_2 B_2 + A_3 B_3)] \\ &\quad + \hat{y} [B_2 (A_1 C_1 + A_2 C_2 + A_3 C_3) - C_2 (A_1 B_1 + A_2 B_2 + A_3 B_3)] \\ &\quad + \hat{z} [B_3 (A_1 C_1 + A_2 C_2 + A_3 C_3) - C_3 (A_1 B_1 + A_2 B_2 + A_3 B_3)] \end{aligned}$$

$$\text{Therefore } \vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$$

1.22

$$(a) \vec{V} = 3\hat{x} - j4\hat{y} + j\hat{z}$$

$$(b) \vec{E} = (3-j4)\hat{x} + 8(1+j)\hat{z}$$

$$(c) \vec{H} = 0.5 e^{-j\pi/4} \hat{x}$$

$$1.23 \quad \frac{\partial}{\partial t} \vec{E}(z, t) \leftrightarrow j\omega \vec{E} = j\omega [(3-j4)\hat{x} + 8(1+j)\hat{z}]$$

1.24 (a) $\vec{C}(t) = \hat{x} \cos \omega t + \hat{y} \sin \omega t$

(b) $\vec{C}(t) = -\hat{x} \sin \omega t + \hat{y} \cos \omega t$

(c) $\vec{C}(t) = \hat{x} \cos(\omega t - k_3) - \hat{y} \sin(\omega t + k_3)$

1.25 (a) $\vec{A} + \vec{B} = (-1+j3)\hat{y} + (1+j3)\hat{z}$

(b) $\vec{A} - \vec{B} = 2\hat{x} + (1-j)\hat{y} + (1+j)\hat{z}$

(c) $\vec{A} \cdot \vec{B} = -1 - (j+2) + (j-2) = -5$

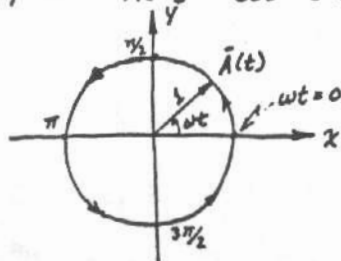
(d) $\vec{A} \times \vec{B} = 4\hat{x} - (1+j3)\hat{y} - (1-j3)\hat{z}$

1.26 $\vec{A} \cdot \vec{A}^* = [\hat{x} + j\hat{y} + (1+j2)\hat{z}] \cdot [\hat{x} - j\hat{y} + (1-j2)\hat{z}] = 1+1+5 = 7$

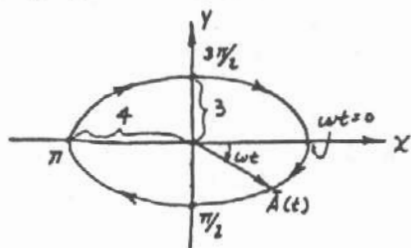
$\vec{A} \times \vec{B}^* = [\hat{x} + j\hat{y} + (1+j2)\hat{z}] \times [-\hat{x} - (1+j2)\hat{y} - j\hat{z}] = (-2+j4)\hat{x} - (1+j)\hat{y} - (1+j)\hat{z}$

$\therefore \text{Re}\{\vec{A} \times \vec{B}^*\} = -2\hat{x} - \hat{y} - \hat{z}$

1.27 (a) $\vec{A} = \hat{x} - j\hat{y} \leftrightarrow \vec{A}(t) = \cos \omega t \hat{x} + \sin \omega t \hat{y}$ and $|\vec{A}(t)| = 1$



(b) $\vec{A} = 4\hat{x} + j3\hat{y} \leftrightarrow \vec{A}(t) = 4\cos \omega t \hat{x} - 3\sin \omega t \hat{y}$



1.28 $\vec{A} = \hat{x} + j2\hat{y}$, $\vec{B} = 2\hat{x} + j\hat{y} \Rightarrow \vec{A} \cdot \vec{B} = 2 - 2 = 0$

But $\vec{A}(t) = \cos \omega t \hat{x} - 2\sin \omega t \hat{y}$ and $\vec{B}(t) = 2\cos \omega t \hat{x} - \sin \omega t \hat{y} \Rightarrow \vec{A}(t) \cdot \vec{B}(t) = 2 - 2\sin^2 \omega t$

$\therefore \vec{A}(t)$ and $\vec{B}(t)$ are not perpendicular at any time.